

HARD rules · blocked by the shield

SOFT rules · penalised in the reward

baseline (QR-DQN)	25%	59%	<1%	6%	27%	62%
+ shield	6%	16%	<1%	19%	33%	41%
+ logic-reward	35%	61%	<1%	5%	12%	37%
+ shield + reward	15%	13%	<1%	15%	32%	23%
+ MCTS	46%	96%	<1%	6%	1%	77%
+ MCTS + shield	40%	26%	<1%	32%	19%	42%
	RG1 safe distance	RG3 speed limit	RI1 no stopping	RG2 no harsh braking	RG4 keep the flow	RI2 no right pass